Muhammad Rafay Waseem ***BS-SE(D)*** ***25I-3166***

Tayyab Ahmed ***BS-SE(D)*** ***25I-3117***

CCP Discussion

Introduction:

The **Dari Mooch** e-commerce project offers premium men's beard grooming products such as beard oils and balms, tailored to different beard types. The target audience is men who wish to maintain a healthy and stylish beard. The website features a clean, modern design that highlights grooming essentials. A unique feature allows users to answer a few quick questions about their beard and skin to discover the perfect product blend. This section explains why the project qualifies as a Complex Computing Problem (CCP).

Problem Complexity Analysis:

The project introduces specific design and technical challenges beyond basic web development. The theme required a tailored user experience for a niche audience, demanding careful attention to layout, responsiveness, and branding. The integration of a personalized quiz feature added complexity, requiring creative problem-solving to design logic that connects user input to product recommendations. Ensuring that the quiz remained lightweight, beginner-friendly, and visually appealing was a significant challenge.

Multiple Solution Approaches Considered:

Several design approaches were considered before finalizing the website. Grid-based layouts, carousel promotions, and modal-based quizzes were explored. The team also debated between using Bootstrap for rapid prototyping and pure CSS for full control. Pure CSS was chosen to maintain brand-specific styling and avoid unnecessary dependencies. For the quiz feature, both form-based and JavaScript-driven implementations were considered. The final solution used vanilla JavaScript to ensure simplicity, performance, and accessibility for future learners.

Technical Challenges Encountered:

During development, alignment and responsiveness issues arose when implementing HTML and CSS. Flexbox and grid systems had to be carefully balanced to achieve consistent layouts across devices. Git conflicts occurred when merging feature branches, which were resolved through structured version control practices and descriptive commit messages. Another challenge was achieving design fidelity when translating Figma mockups into code. Manual adjustments and iterative testing were required to match spacing, fonts, and colors accurately.

Integration of Multiple Technologies:

The project integrates multiple technologies to deliver a cohesive experience. AI-generated images were used for branding and product visuals. HTML and CSS provided the structural and visual foundation. JavaScript powered the interactive quiz and form validations. Git supports version control and collaboration. Together, these components created a professional e-commerce platform. Consistency was ensured through a unified color scheme, modular CSS, and reusable design components across all pages.

Performance and Trade-Off Analysis:

Trade-offs were made between design perfection and development efficiency. Some advanced animations were simplified to improve load times and accessibility. Image optimization was performed to reduce file sizes without compromising quality, ensuring faster performance. The quiz feature added complexity but was scoped carefully to avoid performance bottlenecks. Its logic was written in clean, beginner-friendly JavaScript, balancing functionality with maintainability.

Conclusion:

The **Dari Mooch** e-commerce website qualifies as a Complex Computing Problem because it requires addressing multiple layers of complexity. These included problem-specific design challenges, consideration of alternative solutions, resolution of technical difficulties, integration of diverse technologies, and performance trade-offs. The project demonstrates thoughtful planning, technical skill, and creative execution, making it a strong example of a CCP-worthy implementation.